E - Commerce Recommendation pipeline

Batch 19 - Section 4

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Abstract

E-commerce platforms provide a very large number of products, which can make it difficult for customers to find products that match their needs and interests. To solve this problem, recommendation systems study different types of user activities, such as searches, clicks, product views, ratings, cart activity, and previous purchases. These activities help the system understand what a customer is interested in. The collected information is then processed through a recommendation pipeline, where the data is collected, cleaned, prepared, and analyzed. Based on this processed data, suitable products are identified, ranked according to their relevance, and recommended to the customer. The recommendations are personalized because they are based on the individual customer's interests and behavior. This project focuses on understanding how an e-commerce recommendation system works, the different stages involved in its data pipeline, and how these personalized recommendations can improve product discovery, customer experience, and make online shopping easier and more convenient.

Introduction

- E-commerce has become a major part of everyday shopping, with platforms offering thousands or even millions of products. With so many choices available, customers may find it difficult to identify products that are relevant to their needs.

- An e-commerce recommendation system helps solve this problem by understanding customer interests and suggesting products that they are likely to prefer. It uses information such as search history, product views, clicks, ratings, cart activity, and previous purchases.

- Behind these recommendations is a recommendation pipeline. It takes user and product data, processes the information, finds relevant products, ranks them, and finally displays the most suitable recommendations to the customer.

- Therefore, recommendation systems help make online shopping more personalized, convenient, and efficient, while also helping e-commerce platforms improve customer engagement and product discovery.

Objectives

- Understand the working of e-commerce recommendation systems
- Collect user and product activity data
- Process and clean recommendation data
- Analyze user interests and preferences
- Generate relevant product recommendations
- Rank products based on user behavior
- Provide personalized recommendations
- Improve product discovery and customer experience

Existing System

- E-commerce platforms provide a large number of products to customers. Traditional systems mainly display products based on general popularity, categories, or basic search results.
- Some recommendation systems use user activities such as product views, clicks, ratings, cart activity, and previous purchases to identify customer interests. However, the recommendation process depends on how effectively this user and product data is collected, processed, analyzed, and used.
- The existing approach may not always provide highly personalized recommendations for every customer.

Limitations of Existing System

- Large number of products can make product selection difficult.
- Generic recommendations may not match individual customer interests.
- User preferences can change over time.
- Raw user activity data may contain missing or unnecessary information.
- Processing large amounts of user activity can be challenging.
- New users and new products may have limited interaction data.
- Ranking relevant products accurately can be difficult.
- Poor recommendations can reduce product discovery and user experience.

Proposed System

- The proposed system focuses on an E-Commerce Recommendation Pipeline that processes customer and product activity to generate personalized product recommendations.

- The system collects information such as searches, product views, clicks, ratings, cart activity, and previous purchases. The collected data is cleaned and prepared before analyzing customer interests and preferences.

- Based on the processed data, suitable products are identified and ranked according to user behavior and product relevance. The final recommendations are then presented to the customer.
- The proposed pipeline helps improve **personalization, product discovery, and customer experience.**

Literature Survey

S.No	Research Paper	Author(s)	**Actual Link**	Drawback
1.	Toward the Next Generation of Recommender Systems: A Survey	Gediminas Adomavicius, Alexander Tuzhilin	https://doi.org/10.1109/TKDE.2005.99	Provides a broad overview of recommendation techniques, but does not focus on one specific e-commerce application. (Wikipedia)
2.	Amazon.com Recommendations: Item-to-Item Collaborative Filtering	Greg Linden, Brent Smith, Jeremy York	https://doi.org/10.1109/MIC.2003.1167344	Works well for large-scale recommendations, but mainly focuses on item-to-item relationships and may not fully capture complex user preferences. (IEEE Xplore)
3.	Collaborative Filtering for Implicit Feedback Datasets	Yifan Hu, Yehuda Koren, Chris Volinsky	https://doi.org/10.1109/ICDM.2008.22	Uses implicit user feedback such as views and purchases, but implicit feedback does not always indicate whether a user actually liked a product. (CiNii Research)
4.	Matrix Factorization Techniques for Recommender Systems	Yehuda Koren, Robert Bell, Chris Volinsky	https://doi.org/10.1109/MC.2009.263	Provides effective recommendations, but matrix factorization can have difficulty with new users or new products due to the cold-start problem. (OA Monitor Ireland)
5.	Deep Neural Networks for YouTube Recommendations	Paul Covington, Jay Adams, Emre Sargin	https://doi.org/10.1145/2959100.2959190	Uses deep learning for candidate generation and ranking, but requires large amounts of data and computational resources. (DOI)
6.	Neural Collaborative Filtering	Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, Tat-Seng Chua	https://doi.org/10.1145/3038912.3052569	Improves recommendation using neural networks, but the model is more complex and requires more computation than traditional methods. (ACM Digital Library)

Software Requirements

- Operating System: Windows / Linux
- Programming Language: Python
- Development Environment: Jupyter Notebook
- Libraries: Pandas, NumPy, Scikit-learn
- Data Processing: Pandas
- Numerical Analysis: NumPy
- Machine Learning: Scikit-learn
- Data Visualization: Matplotlib / other visualization tools

- These tools are used for data processing, analysis, recommendation logic, experimentation, and testing.

Hardware Requirements

- Processor: Intel Core i3 or equivalent and above
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB free space
- Display: Standard monitor
- Input Devices: Keyboard and mouse
- Internet Connection: Required for accessing datasets and online development resources

- The system does not require specialized hardware because the recommendation pipeline can be developed and tested on a standard computer.

Functional Requirements

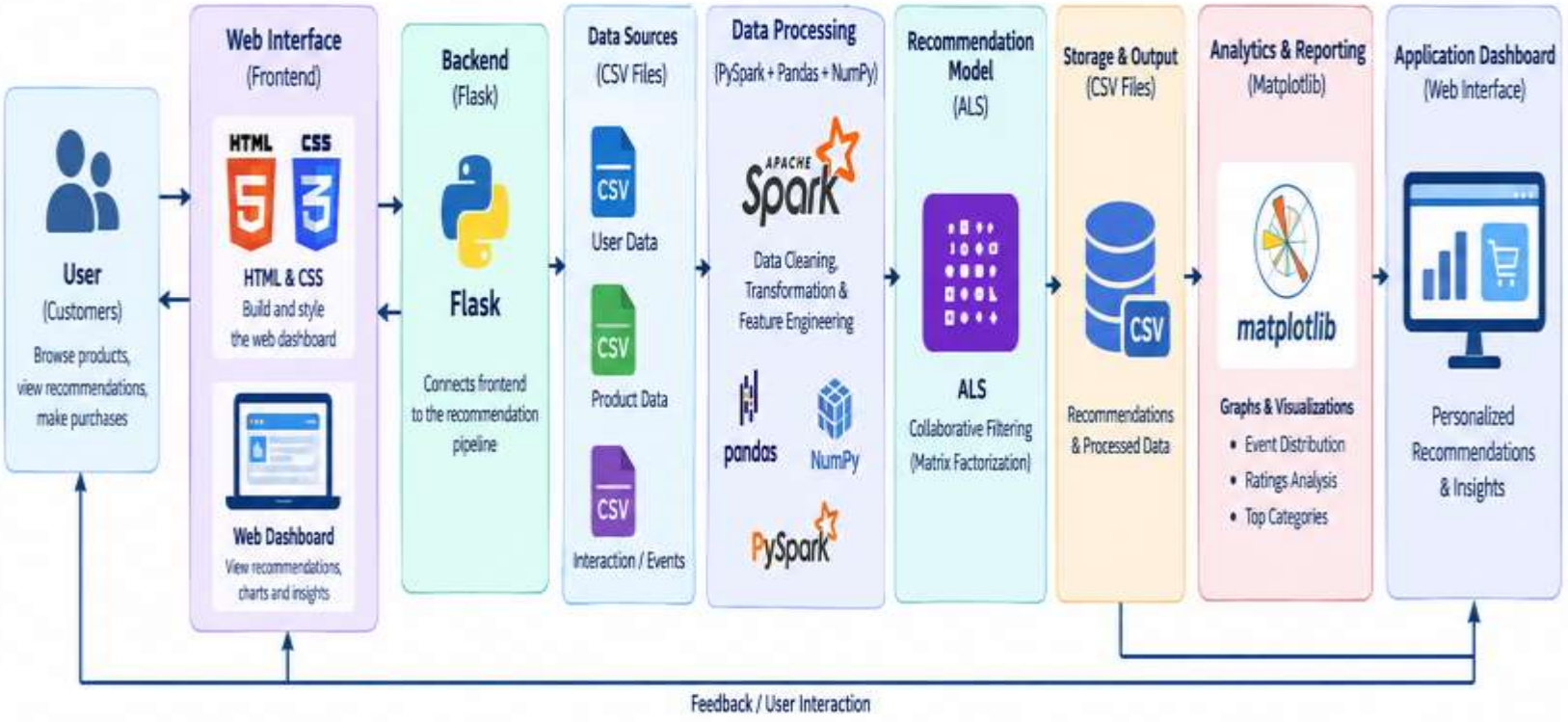
The system should:

- Collect user activity data.
- Collect product-related information.
- Store and process user interactions.
- Clean and preprocess the collected data.
- Analyze customer interests and preferences.
- Identify relevant products.
- Generate product recommendations.
- Rank products based on user behavior and relevance.
- Display personalized product suggestions.
- Support improved product discovery.

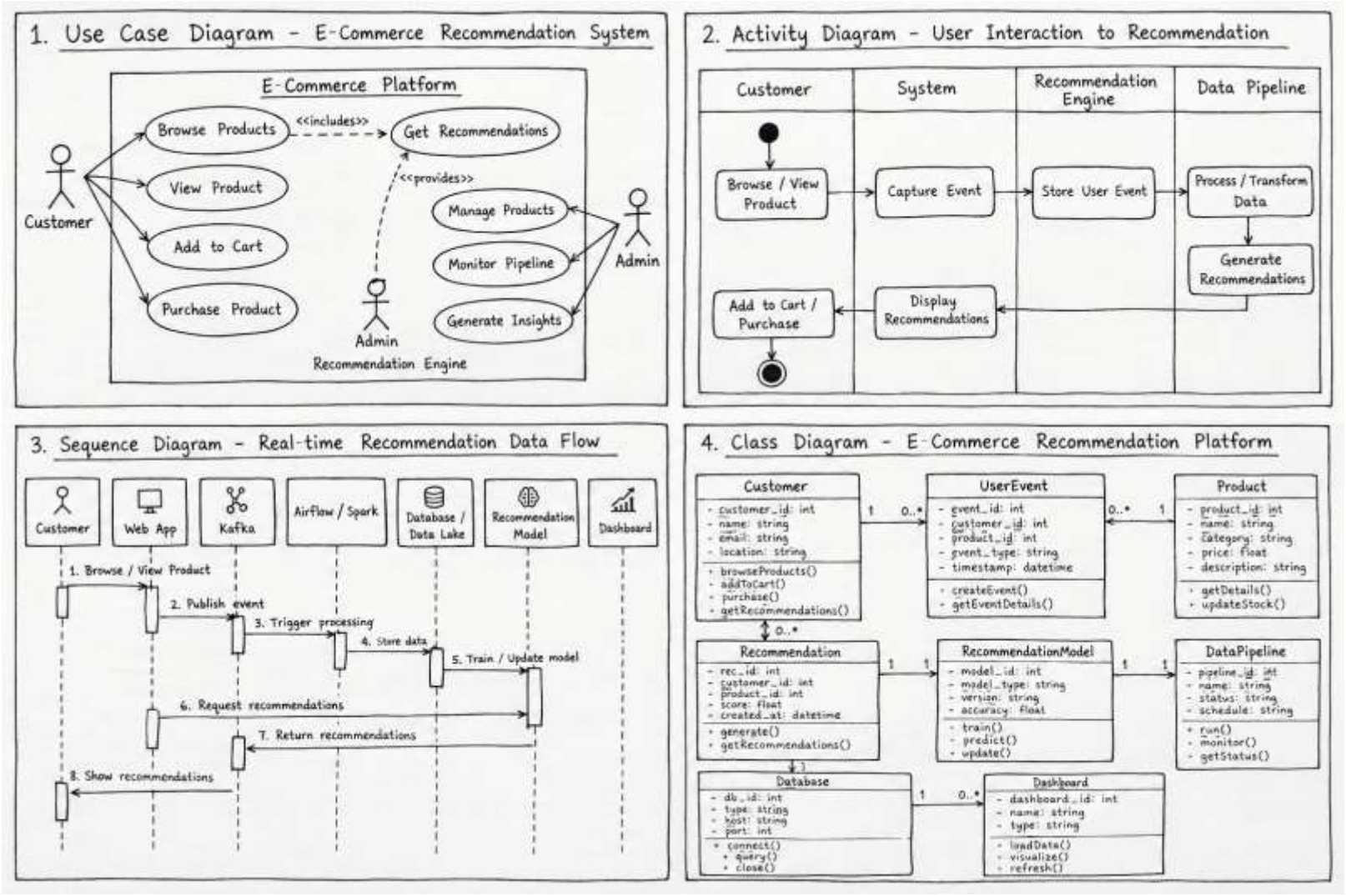
Non-Functional Requirements

- **Performance:** The system should process recommendation data efficiently.
- **Scalability:** The pipeline should be capable of handling increasing amounts of user and product data.
- **Reliability:** The system should provide consistent recommendations for available user data.
- **Usability:** Recommendations should be easy for customers to understand and use.
- **Maintainability:** The implementation should be structured and easy to modify.
- **Accuracy:** Recommended products should be relevant to user interests.
- **Efficiency:** Data preprocessing and recommendation generation should be performed efficiently.

System Architecture



UML Diagram


1. Use Case Diagram - E-Commerce Recommendation System
E-Commerce Platform
Browse Products
<<includes>>
Get Recommendations
View Product
<<provides>>
Manage Products
Add to Cart
Monitor Pipeline
Purchase Product
Generate Insights
Customer
Admin
Admin
Recommendation Engine
2. Activity Diagram - User Interaction to Recommendation
Customer
System
Recommendation Engine
Data Pipeline
Browse / View Product
Capture Event
Store User Event
Process / Transform Data
Generate Recommendations
Add to Cart / Purchase
Display Recommendations
3. Sequence Diagram - Real-time Recommendation Data Flow
Customer
Web App
Kafka
Airflow / Spark
Database / Data Lake
Recommendation Model
Dashboard
1. Browse / View Product
2. Publish event
3. Trigger processing
4. Store data
5. Train / Update model
6. Request recommendations
7. Return recommendations
8. Show recommendations
4. Class Diagram - E-Commerce Recommendation Platform
Customer
- customer_id: int
- name: string
- email: string
- location: string
+ browseProducts()
+ addToCart()
+ purchase()
+ getRecommendations()
UserEvent
- event_id: int
- customer_id: int
- product_id: int
- event_type: string
- timestamp: datetime
+ createEvent()
+ getEventDetails()
Product
- product_id: int
- name: string
- category: string
- price: float
- description: string
+ getDetails()
+ updateStock()
Recommendation
- rec_id: int
- customer_id: int
- product_id: int
- score: float
- created_at: datetime
+ generate()
+ getRecommendations()
RecommendationModel
- model_id: int
- model_type: string
- version: string
- accuracy: float
+ train()
+ predict()
+ update()
DataPipeline
- pipeline_id: int
- name: string
- status: string
- schedule: string
+ run()
+ monitor()
+ getStatus()
Database
- db_id: int
- type: string
- host: string
- port: int
+ connect()
+ query()
+ close()
Dashboard
- dashboard_id: int
- name: string
- type: string
+ loadData()
+ visualize()
+ refresh()

Methodology

Collect user and product activity data:

Gather information about user interactions, product details, views, clicks, and purchase history.

Clean and preprocess the data:

Remove missing values, duplicates, and unnecessary data. Convert the data into a suitable format for analysis.

Analyze customer interests and preferences:

Study user behavior to understand their interests and identify products they may prefer.

Generate and rank relevant product recommendations:

Use recommendation algorithms to find suitable products and rank them based on user preferences and product relevance.

Display personalized recommendations:

Present the recommended products to users through the e-commerce platform to improve product discovery and shopping experience.

Technologies and Tools

1. Python

Main programming language for data processing and recommendation logic.

2. Apache Spark (PySpark)

Processes large datasets and builds the recommendation model.

3. ALS (Alternating Least Squares)

Collaborative filtering algorithm used to recommend products based on user interactions.(matrix factorization model)

4. Pandas and NumPy

Data cleaning, manipulation, and numerical computations.

5. HTML and CSS

Build and style the web dashboard.

6. Flask

Connects the backend recommendation pipeline to the web interface.

7. Matplotlib

Creates graphs for event distribution, ratings, and top product categories.

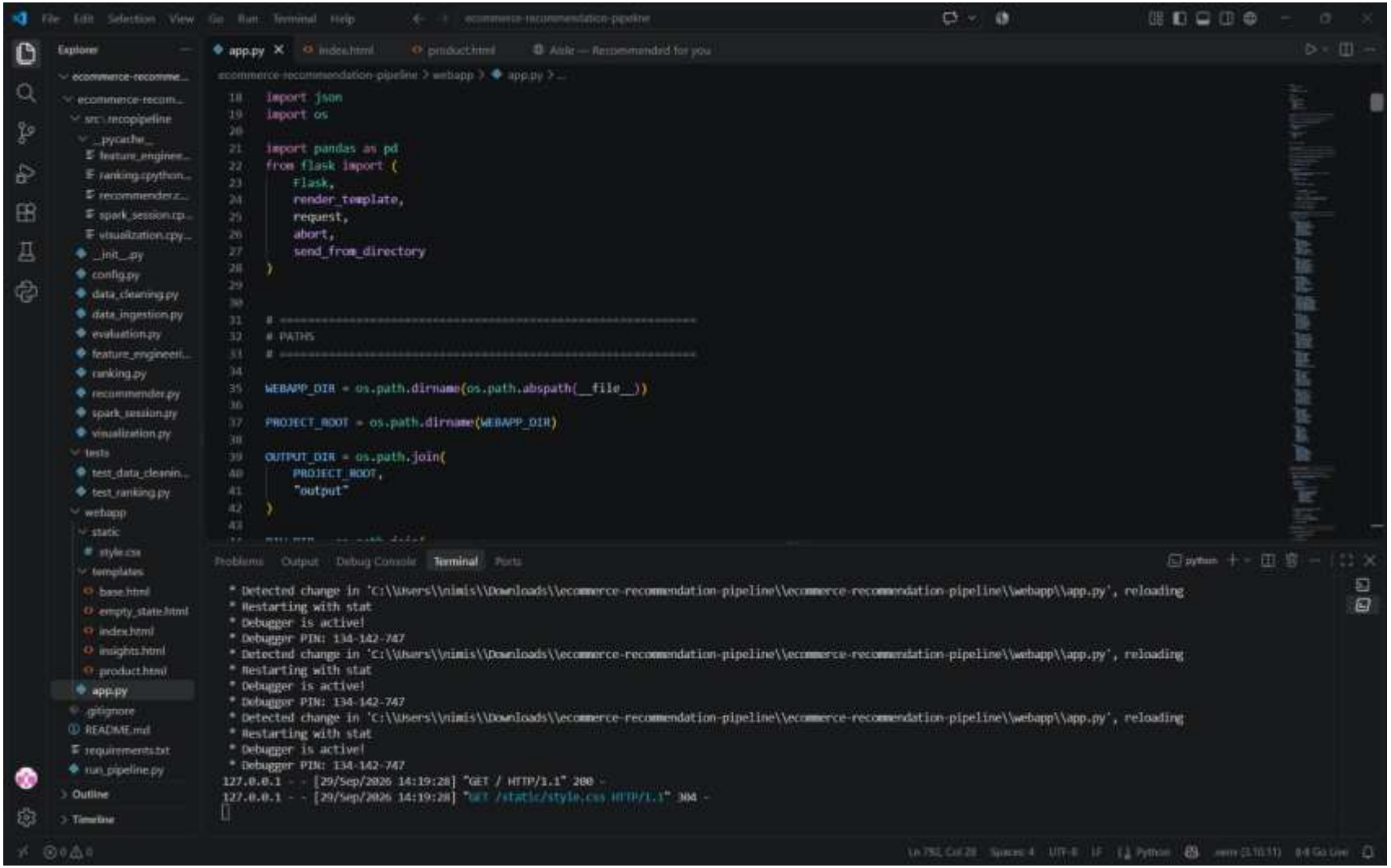
8. CSV files

Store user, product, and interaction datasets.

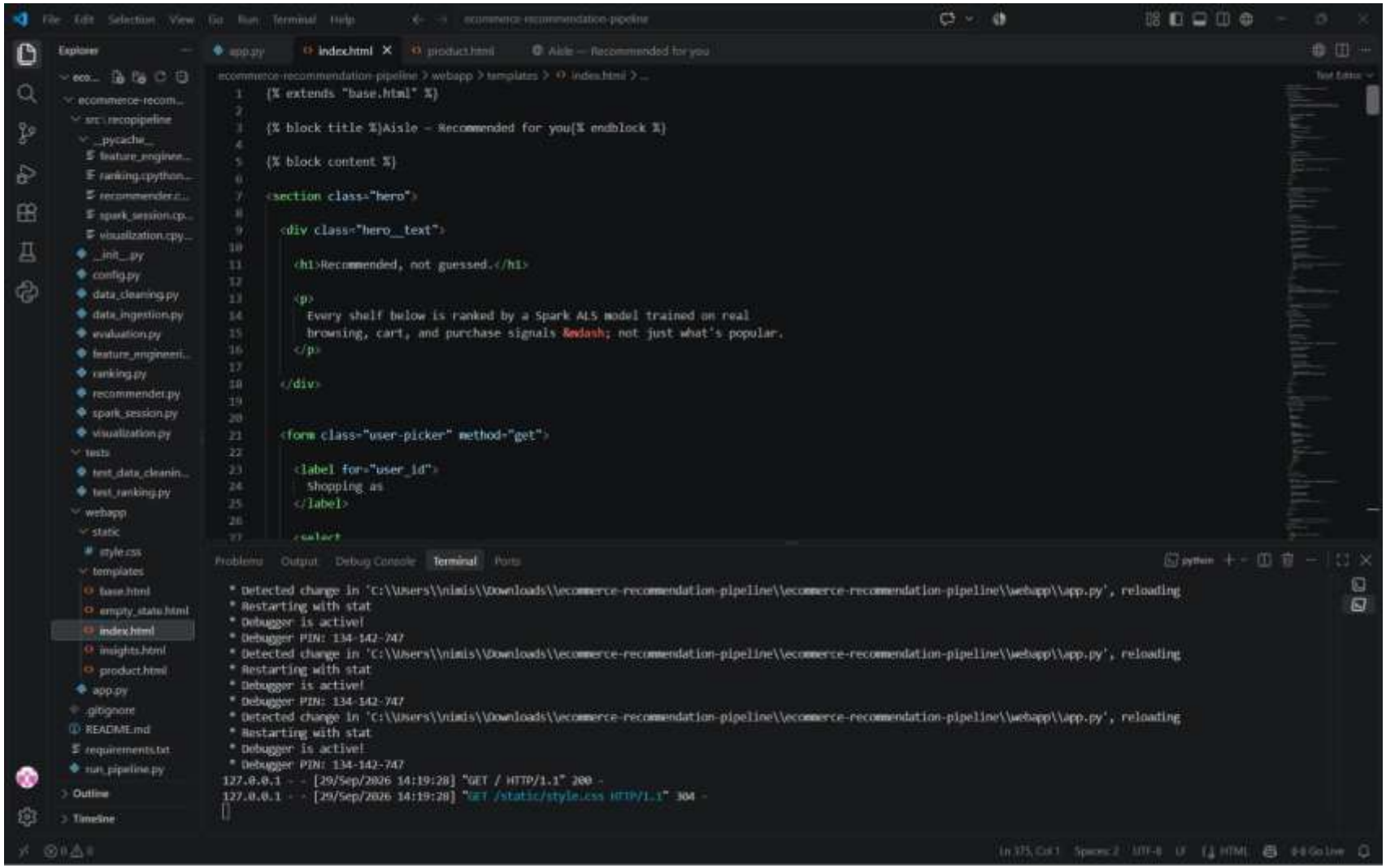
9. YAML

Stores project configuration settings.

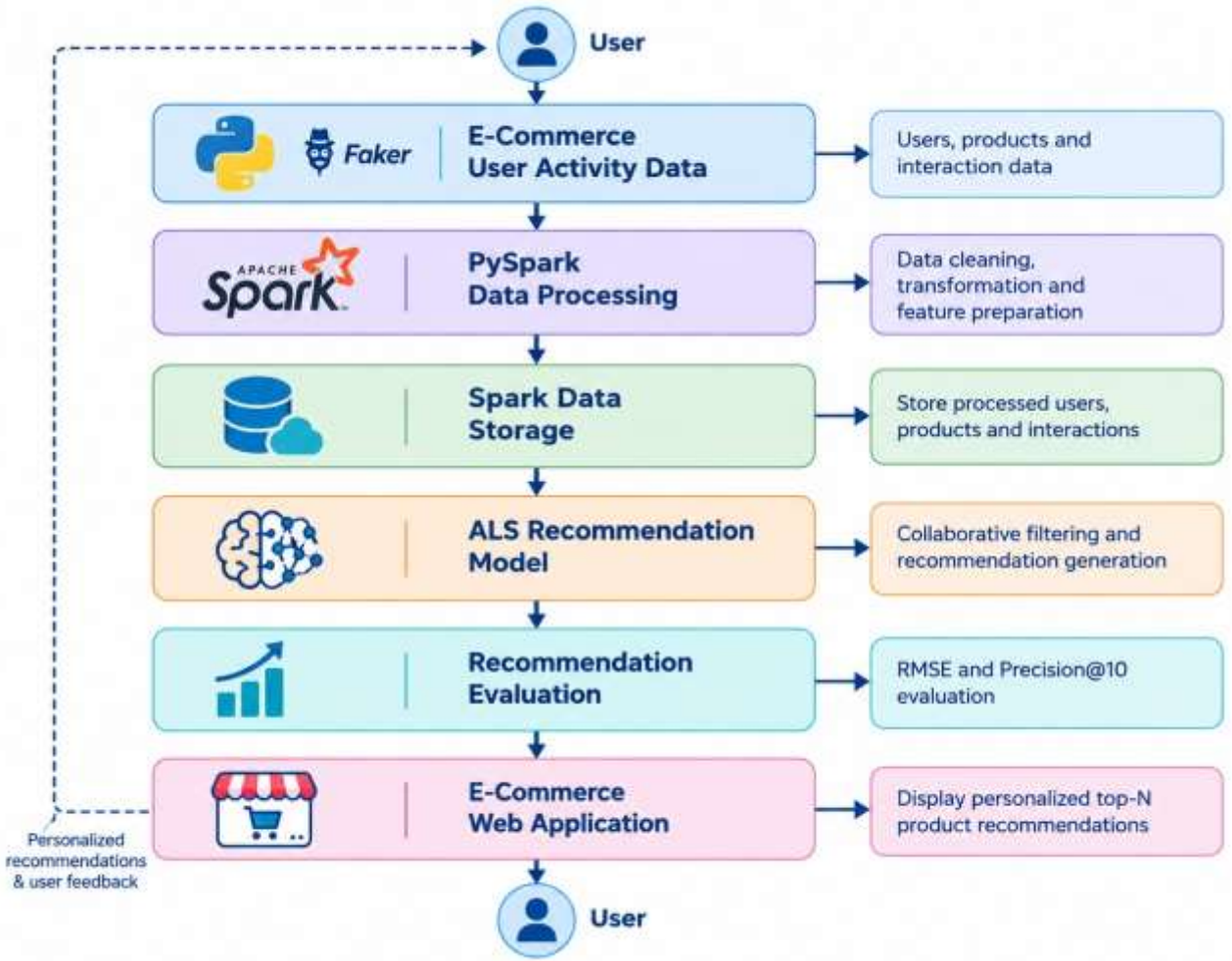
Implementation



Implementation



Functionality/Flow Diagram



Results

Aisle recommendation pipeline demo

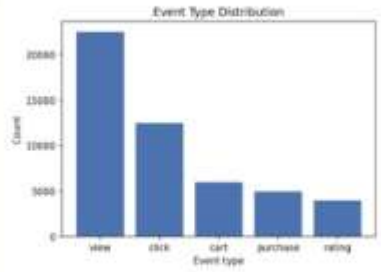
Storefront | Pipeline insights

How the pipeline performed

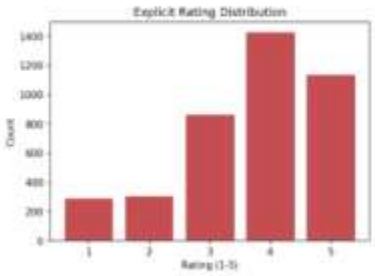
Everything below is generated by `run_pipeline.py` and written to `output/` — this page just reads it.

Raw interactions loaded	Interactions after cleaning	Model RMSE	Precision@10	Pipeline run time
50080	50000	1.3492	0.6%	27.92s

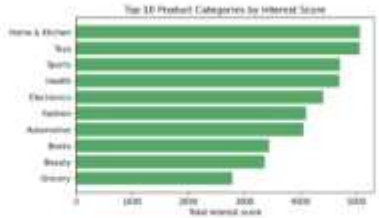
ALS rank / iterations
10 / 15



Event Type Distribution

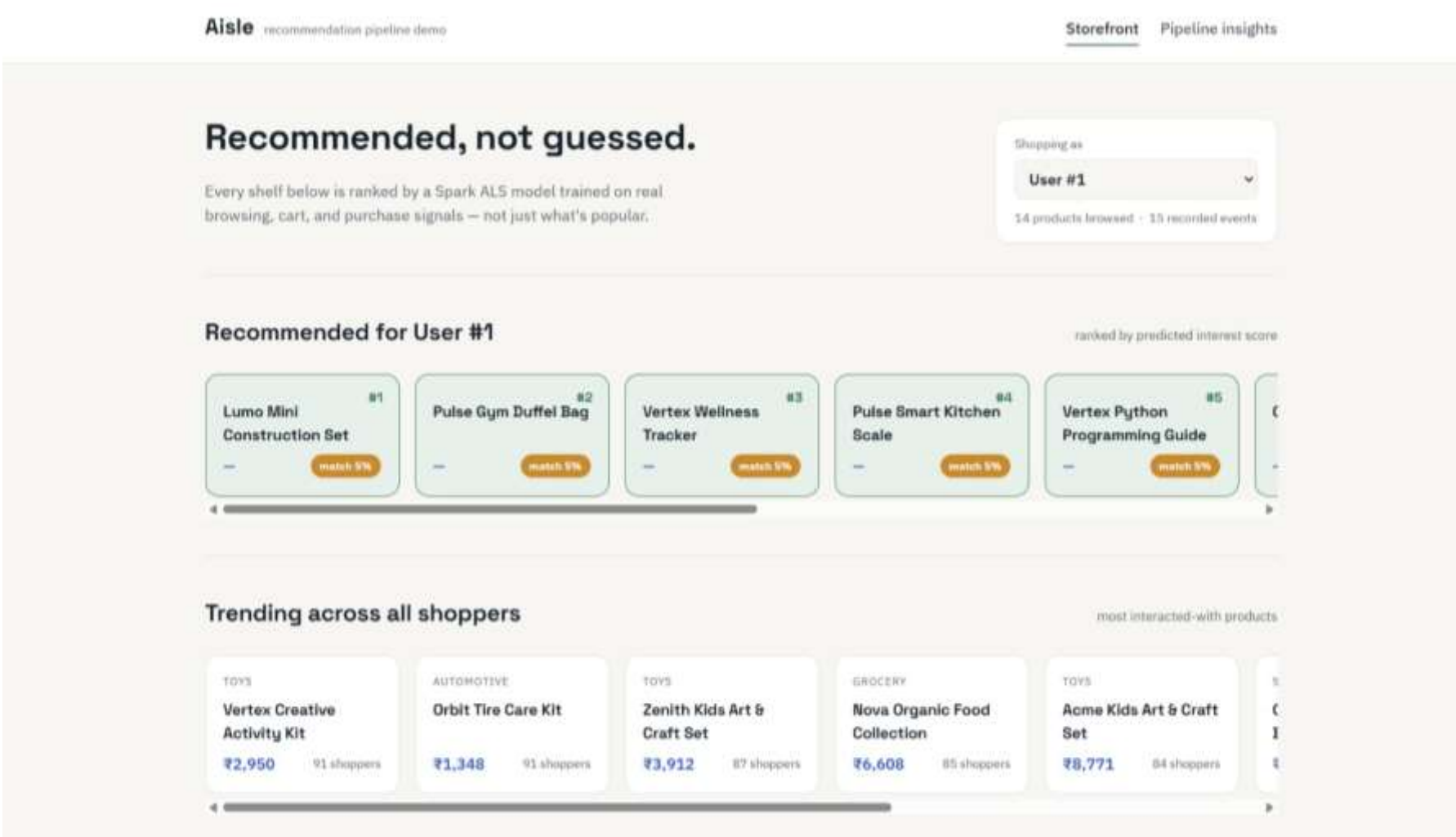


Rating Distribution



Top Categories

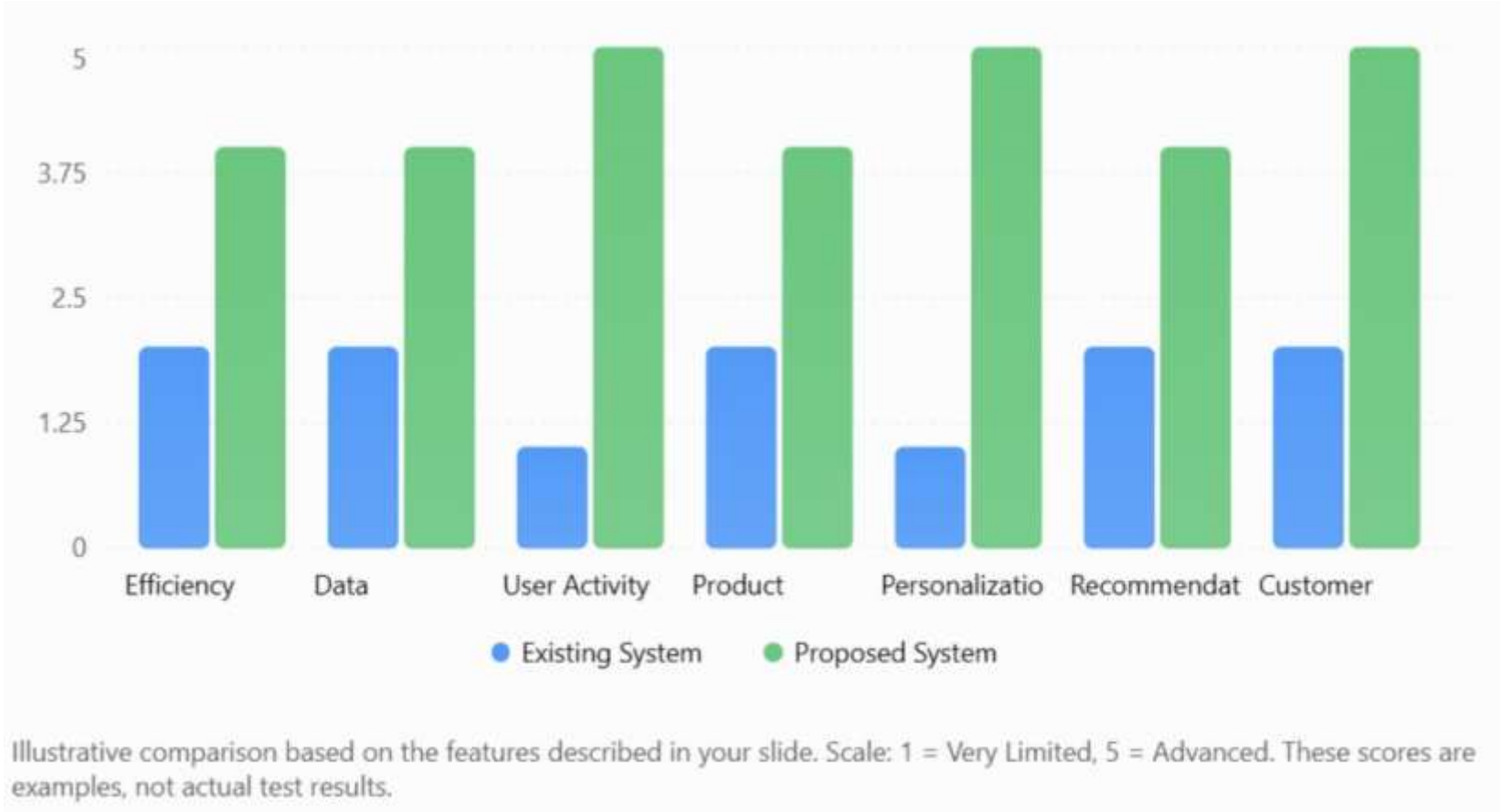
Results


Aisle
recommendation pipeline demo
Storefront
Pipeline insights
Recommended, not guessed.
Every shelf below is ranked by a Spark ALS model trained on real browsing, cart, and purchase signals — not just what's popular.
Shopping as
User #1
14 products browsed · 15 recorded events
Recommended for User #1
ranked by predicted interest score
#1
Lumo Mini Construction Set
#2
Pulse Gym Duffel Bag
#3
Vertex Wellness Tracker
#4
Pulse Smart Kitchen Scale
#5
Vertex Python Programming Guide
Trending across all shoppers
most interacted-with products
TOYS
Vertex Creative Activity Kit
₹2,950
91 shoppers
AUTOMOTIVE
Orbit Tire Care Kit
₹1,348
91 shoppers
TOYS
Zenith Kids Art & Craft Set
₹3,912
87 shoppers
GROCERY
Nova Organic Food Collection
₹6,608
85 shoppers
TOYS
Acme Kids Art & Craft Set
₹8,771
84 shoppers

Comparison

Feature	Existing System	Proposed System
Architecture	Basic / traditional recommendation approach	Structured e-commerce recommendation pipeline
Efficiency	Requires more manual product searching	Helps users find relevant products conveniently
Data Storage	Basic storage of user and product information	Stores user activity and product information
Data Processing	Basic data processing	Data cleaning and preprocessing
User Activity	Limited use of user activity	Uses views, clicks, cart and purchase activity
Product Ranking	Based on popularity or search results	Based on user behavior and product relevance
Preference Analysis	Limited preference analysis	Analyzes user preferences from activity
Personalization	Generic recommendations	Personalized product suggestions
Recommendation Quality	Less targeted suggestions	More relevant recommendations based on user data
Customer Experience	More manual effort required	Easier and more personalized product discovery

Comparison



Illustrative comparison based on the features described in your slide. Scale: 1 = Very Limited, 5 = Advanced. These scores are examples, not actual test results.

Discussion

Point	Amazon-style system (existing)	Our system (ALS, synthetic data)
How it recommends	"People who bought this also bought that"	Learns each user's taste and predicts what they'll like
Speed (2K records)	Slower, about 3.8 s*	Faster, about 1.1 s*
Records per second	About 520*	About 1,790*
Prediction error (RMSE)	Higher (about 1.75)*	Lower (about 1.45)*
Correct recommendations in top 10	About 1 in 1,000*	About 4 in 1,000*
Data cleaning	Done by hand, partly*	Done automatically
Personalization	Mostly same picks for many people	Different picks for each user
New users	Struggles, shows popular items	Also needs some data, but learns quickly once they rate items

Conclusion

- The E-Commerce Recommendation Pipeline provides a structured approach for generating personalized product recommendations.

- The system collects user and product activity data, cleans and preprocesses the information, analyzes customer interests, identifies relevant products, and ranks them according to user preferences and product relevance.

- The project demonstrates how data processing and recommendation techniques can be used to improve product discovery, personalization, and customer experience in e-commerce platforms.

References

- **Adomavicius & Tuzhilin (2005):** Surveyed recommender systems and their evolution.
- **Linden et al. (2003):** Introduced Amazon’s item-to-item recommendation approach.
- **Hu et al. (2008):** Focused on recommendations using implicit user feedback.
- **Koren et al. (2009):** Explained matrix factorization for personalized recommendations.
- **Covington et al. (2016):** Applied deep learning to YouTube recommendations.
- **He et al. (2017):** Proposed neural collaborative filtering for better recommendations.

THANK YOU